

CHAPTER 7

DIFFERENCE EQUATIONS

7.1 DIFFERENCE EQUATIONS

A difference equation of order N is an equation involving differences $y_n = \Delta^0 y_n, \Delta y_n, \Delta^2 y_n, \dots, \Delta^N y_n$ of a number sequence i.e.

$$\Delta^N y_n = f(n, y_n, \Delta y_n, \Delta^2 y_n, \dots, \Delta^{N-1} y_n). \quad (7.1)$$

Since $\Delta y_n = y_{n+1} - y_n$,

$$\begin{aligned} \Delta^2 y_n &= \Delta(\Delta y_n) = \Delta(y_{n+1} - y_n) = \Delta y_{n+1} - \Delta y_n \\ &= y_{n+2} - y_{n+1} - y_{n+1} + y_n = y_{n+2} - 2y_{n+1} + y_n \end{aligned}$$

etc., it is possible to express equation (7.1) in terms of $y_n, y_{n+1}, \dots, y_{n+N}$. If it is possible to write the equation in the form

$$y_{n+N} + c_1 y_{n+N-1} + \dots + c_N y_n = g_n \quad (7.2)$$

where $c_1, c_2, \dots, c_N$ and g_n may depend on n but not on $y_n, y_{n+1}, \dots, y_{n+N}$, then the equation is said to be a linear difference equation of order N . If the coefficients $c_1, c_2, \dots, c_N$ depend on n then (7.2) is called a linear difference equation of order N with variable coefficients otherwise it is called a linear difference equation of order N with constant coefficients.

A solution of a difference equation consists in finding a

sequence $\{y_n\}_{n=0}^{\infty}$ such that $y_n, y_{n+1}, \dots, y_{n+N}$ substituted in (7.2) satisfy it.

If the equation is written in the form (7.2) its order is easily seen to be the difference between the highest and the lowest subscript. For example, the orders of difference equations given in examples (7.1), (7.2), (7.5) are respectively 1, 1, and 2.

EXAMPLE 7.1: Solve the initial value problem

$$y_{n+1} - 2y_n = 1, \quad y_0 = 1.$$

SOLUTION: We have $y_1 = 2y_0 + 1 = 2 + 1 = 3 = 2^2 - 1$

$$y_2 = 2y_1 + 1 = 2 \cdot 3 + 1 = 7 = 2^3 - 1$$

$$y_3 = 2y_2 + 1 = 2 \cdot 7 + 1 = 15 = 2^4 - 1$$

or

$$y_n = 2^{n+1} - 1$$

is the solution of the given difference problem.

EXAMPLE 7.2: Solve the initial value problem

$$y_{n+1} = r y_n + n, \quad y_0 = a.$$

SOLUTION: Here $y_1 = r y_0 = ra$

$$y_2 = r y_1 + 1 = r(ra) + 1 = r^2 a + 1$$

$$y_3 = r y_2 + 2 = r(r^2 a + 1) + 2 = r^3 a + r + 2$$

$$y_4 = r y_3 + 3 = r(r^3 a + r + 2) + 3 = r^4 a + r^2 + 2r + 3$$

or

$$y_n = r^n a + r^{n-2} + 2r^{n-3} + 3r^{n-4} + \dots + n-1.$$

EXAMPLE 7.3: Find the difference equation for

$$I_n = \int_0^{\pi/4} \tan^n x \, dx,$$

and then solve it.

SOLUTION:
$$I_n = \int_0^{\pi/4} \tan^n x \, dx = \int_0^{\pi/4} \tan^{n-2} x (\sec^2 x - 1) dx$$

$$\begin{aligned}
 I_n &= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx - I_{n-2} \\
 &= \left[\frac{\tan^{n-1} x}{n-1} \right]_0^{\pi/4} - I_{n-2} = \frac{1}{n-1} - I_{n-2}
 \end{aligned}$$

or
$$I_{n+1} + I_{n-1} = 1/n, \quad n = 1, 2, \dots$$

Also
$$I_0 = \pi/4, \quad I_1 = \left[-\ln \cos x \right]_0^{\pi/4} = \frac{1}{2} \ln 2.$$

Thus
$$\begin{aligned}
 I_2 &= 1 - I_0, \\
 I_4 &= 1/3 - I_2 = 1/3 - 1 + I_0, \\
 I_6 &= 1/5 - 1/3 + 1 - I_0,
 \end{aligned}$$

and so on.

For the odd subscript, we have

$$\begin{aligned}
 I_3 &= 1/2 - I_1 \\
 I_5 &= 1/4 - 1/2 + I_1 \\
 I_7 &= 1/6 - 1/4 + 1/2 - I_1,
 \end{aligned}$$

and so on.

EXAMPLE 7.4: Find the number of years required for a given money to triple at (a) compound interest rate 10 % per annum.
(b) simple interest rate 10 % per annum.

SOLUTION: a) Let y_0 be the initial deposit and let y_k be the sum on deposit after year number k . Then the difference equation is

$$\begin{aligned}
 y_{k+1} &= y_k + ry_k, \quad r = 10/100 = 1/10 \\
 &= (1 + r)y_k
 \end{aligned}$$

Now

$$\begin{aligned}
 y_1 &= (1 + r)y_0 \\
 y_2 &= (1 + r)y_1 = (1 + r)^2 y_0 \\
 y_3 &= (1 + r)y_2 = (1 + r)^3 y_0.
 \end{aligned}$$

Thus the general solution of the difference equation is

$$y_k = (1 + r)^k y_0.$$

If the amount becomes triple after k years, then

$$\begin{aligned} 3y_0 &= (1+r)^k y_0 \\ k \ln(1+r) &= \ln 3 \\ k &= \frac{\ln 3}{\ln(1+r)} = \frac{\ln 3}{\ln(1.1)} = 11.53. \end{aligned}$$

Thus the amount will become triple after 11.53 years.

b) Here $y_{k+1} = y_k + ry_0$, $r = 10/100 = 1/10$

$$\begin{aligned} y_1 &= y_0 + ry_0 = (1+r)y_0 \\ y_2 &= y_1 + ry_0 = (1+r)y_0 + ry_0 = (1+2r)y_0 \\ y_3 &= y_2 + ry_0 = (1+2r)y_0 + ry_0 = (1+3r)y_0 \end{aligned}$$

Thus the general solution of this difference equation is

$$y_k = (1+kr)y_0.$$

If the amount will become triple after k years, then

$$\begin{aligned} 3y_0 &= (1+kr)y_0 \\ kr &= 2 \quad \text{or } k = 2/r = 20. \end{aligned}$$

Hence the amount becomes triple after 20 years.

7.2 LINEAR HOMOGENEOUS DIFFERENCE EQUATIONS

We shall treat only difference equations with constant coefficients. First we consider the equation

$$y_n + c_1 y_{n-1} + \dots + c_{n-1} y + c_n y_0 = 0. \quad (7.3)$$

Such an equation with zero on the right hand side is said to be homogeneous. To find a solution, we assume that $y_n = \alpha^n$ where α is a constant to be determined. Substituting in (7.3) we find that $y_n = \alpha^n$ will be a solution if

$$\alpha^n + c_1 \alpha^{n-1} + \dots + c_{n-1} \alpha + c_n = 0. \quad (7.4)$$

This is called the characteristic equation of the difference equation (7.3). Three cases arise according as the roots of (7.4) are

- a) real and distinct
- b) complex
- c) real and repeated

a) **Real and distinct roots.** Suppose $\alpha_1, \alpha_2, \dots, \alpha_n$ are distinct roots of the characteristic equation (7.4). Then $\alpha_1^n, \alpha_2^n, \dots, \alpha_n^n$ are all solutions of the difference equation (7.3) and since the equation is linear, any linear combination of these solutions will also be a solution.

$$\text{Thus } y_n = a_1(\alpha_1)^n + a_2(\alpha_2)^n + \dots + a_n(\alpha_n)^n, \quad (7.5)$$

where $a_1, a_2, \dots, a_n$ are arbitrary constants, is a solution of the given difference equation. This is called the general solution of the equation.

EXAMPLE 7.5: Solve $y_{n+2} - 5y_{n+1} + 4y_n = 0$.

SOLUTION: Let $y_n = \alpha^n$. Then the characteristic equation is

$$\alpha^2 - 5\alpha + 4 = 0;$$

with roots 4 and 1. Hence the general solution is

$$\begin{aligned} y_n &= a_1 4^n + a_2 1^n \\ &= a_1 4^n + a_2. \end{aligned}$$

EXAMPLE 7.6: Solve $y_{n+3} + y_{n+2} - 4y_{n+1} - 4 = 0$.

SOLUTION: The characteristic equation is

$$\begin{aligned} \alpha^3 + \alpha^2 - 4\alpha - 4 &= 0, \\ \text{or } (\alpha + 1)(\alpha^2 - 4) &= 0. \end{aligned}$$

Thus $-1, \pm 2$ are roots of the characteristic equation. Hence the general solution is

$$y_n = a_1(-1)^n + a_2(2)^n + a_3(-2)^n.$$

b) Complex roots. If the characteristic equation has a pair of complex conjugate roots

$$\alpha_1 = a + ib$$

$$\alpha_2 = a - ib$$

we express them in polar form

$$\alpha_1 = re^{i\phi}$$

$$\alpha_2 = re^{-i\phi}$$

Since $(\alpha_1)^n$ and $(\alpha_2)^n$ satisfy the equation, we have two solutions

$$u_1 = r^n e^{in\phi} \quad \text{and} \quad u_2 = r^n e^{-in\phi}.$$

Since the equation is linear, $\frac{u_1 + u_2}{2}$ and $\frac{u_1 - iu_2}{2i}$ are also solutions. But

$$\frac{u_1 + u_2}{2} = r^n \frac{e^{in\phi} + e^{-in\phi}}{2} = r^n \cos n\phi,$$

$$\text{and} \quad \frac{u_1 - iu_2}{2i} = r^n \frac{e^{in\phi} - ie^{-in\phi}}{2i} = r^n \sin n\phi.$$

Thus $r^n \cos n\phi$ and $r^n \sin n\phi$ are two linearly independent solutions corresponding to the pair of roots $a \pm ib$ of the characteristic equation. Here (r, ϕ) are the polar co-ordinates of the complex number $a \pm ib$.

EXAMPLE 7.7: Solve the difference equation $y_{n+2} + y_n = 0$.

SOLUTION: The characteristic equation is $\alpha^2 + 1 = 0$ with roots $\pm i$.

Since $i = e^{i\pi/2}$,

we find $\cos(n\pi/2)$ and $\sin(n\pi/2)$ as two independent solutions of the difference equation and the general solution is

$$y_n = a_1 \cos(n\pi/2) + a_2 \sin(n\pi/2).$$

EXAMPLE 7.8: Solve $y_{n+3} - 5y_{n+2} + 8y_{n+1} - 6y_n = 0$.

SOLUTION: The characteristic equation is

$$\alpha^3 - 5\alpha^2 + 8\alpha - 6 = (\alpha - 3)(\alpha^2 - 2\alpha + 2) = 0.$$

The roots are $3, -1 \pm i$.

Now $-1 + i = \sqrt{2} e^{3\pi i/4}$,

therefore two solutions corresponding to the roots $-1 \pm i$, of the characteristic equation, are

$$(2)^{n/2} \cos \frac{3\pi n}{4} \text{ and } (2)^{n/2} \sin \frac{3\pi n}{4}.$$

Hence the general solution may be written as

$$y_n = a_1 3^n + a_2 2^{n/2} \cos \frac{3\pi n}{4} + a_3 2^{n/2} \sin \frac{3\pi n}{4}$$

where a_1, a_2, a_3 are arbitrary constants.

c) Real and repeated roots. Consider the equation

$$y_{n+2} - 2\beta y_{n+1} + \beta^2 y_n = 0.$$

The characteristic equation, $\alpha^2 - 2\alpha\beta + \beta^2 = 0$, has equal roots β, β . In this case $(\beta)^n$ is a solution of the difference equation. We find by direct substitution that $n(\beta)^n$ is another, independent solution. Hence, in this case we have the general solution

$$y_n = a_1 \beta^n + a_2 n\beta^n.$$

EXAMPLE 7.9: Suppose we wish to solve a difference equation whose characteristic equation is of the form

$$(\alpha^2 - 2\alpha + 3)(\alpha^2 + 4\alpha + 4) = 0.$$

The roots are $1 \pm \sqrt{2} i$ and -2 repeated twice.

Since $1 + \sqrt{2} i = \sqrt{3} e^{i\phi}$, where $\phi = \tan^{-1}\sqrt{2}$,

two solutions corresponding to the pair of complex roots $1 \pm \sqrt{2} i$

are $3^{n/2} \cos n\phi$ and $3^{n/2} \sin n\phi$. Two other solutions corresponding to the repeated roots -2 are $(-2)^n$, and $n(-2)^n$. Hence the general solution may be written as

$$y_n = a_1(-2)^n + a_2 n(-2)^n + a_3 3^{n/2} \cos n\phi + a_4 3^{n/2} \sin n\phi.$$

7.3 LINEAR INHOMOGENEOUS DIFFERENCE EQUATIONS

Let us now consider equations of the type

$$y_n + c_1 y_{n-1} + \dots + c_k y_{n-k} = g_n \quad (k = 1, 2, \dots, n) \quad (7.6)$$

with $g_n \neq 0$. Such equations are called inhomogeneous equations. Suppose we can find a $y_n = u_n$ which satisfies (7.6). The general solution is found by adding u_n to the general solution of the corresponding homogeneous equation obtained from (7.6) by setting $g_n = 0$.

We shall consider only simple cases in which g_n is a polynomial in n , an exponential function or is a sinusoidal function of n . The method is really a hit and trial method and essentially depends on making an intelligent guess about the solution u_n which is called the particular solution of the equation. However, the following table will be helpful for the trial solution.

g_n	Trial solution
a^n	$c_1 a^n$
n^k	$c_0 + c_1 n + c_2 n^2 + \dots + c_k n^k$
$\sin bn$ or $\cos bn$	$c_0 \sin bn + c_1 \cos bn$
$\alpha^n \sin bn$ or $\alpha^n \cos bn$	$\alpha^n (c_0 \sin bn + c_1 \cos bn)$

If the trial solution includes some terms of the solution of associated homogeneous equation of (7.6) i.e., the corresponding homogeneous equation obtained from (7.6) by setting $g_n = 0$, then the trial solution will be multiplied by a suitable power of k .

EXAMPLE 7.10: Solve the difference equation

$$y_n + 3y_{n-1} + 2y_{n-2} = 2.$$

SOLUTION: The corresponding homogeneous equation is

$$y_n + 3y_{n-1} + 2y_{n-2} = 0,$$

whose solution is $y_n = a_1(-1)^n + a_2(-2)^n$.

Let us now find the particular solution. Assume $y_n = c$ satisfies the given difference equation. This will be so if $c + 3c + 2c = 2$ or $c = 1/3$. Thus the general solution is

$$y_n = a_1(-1)^n + a_2(-2)^n + 1/3.$$

EXAMPLE 7.11: Find a particular solution of the equation

$$y_n - 2y_{n-1} + 3y_{n-2} = 2n^2 - 3n + 4.$$

SOLUTION: We assume $u_n = An^2 + Bn + C$, then

$$u_{n-1} = A(n-1)^2 + B(n-1) + C$$

$$u_{n-2} = A(n-2)^2 + B(n-2) + C.$$

If u_n is to satisfy the given equation, then for all n :

$$(A-2A+3A)n^2 + (4A-12A+B-2B+3B)n - 2A+12A+2B-6B+C-2C+3C = 2n^2 - 3n + 4.$$

Comparing coefficients of n^2 , n and the constant terms on both sides we have

$$2A = 2 \Rightarrow A = 1$$

$$-8A + 2B = -3 \Rightarrow 2B = -3 + 8A \Rightarrow B = 5/2$$

$$10A - 4B + 2C = 4 \Rightarrow C = 2.$$

Hence

$$u_n = n^2 + (5/2)n^2 + 2,$$

is a particular solution of the equation.

EXAMPLE 7.12: Find a particular solution of the equation

$$y_n - 3y_{n-1} + 2y_{n-2} = n^2 - 2n + 8.$$

SOLUTION: Here the substitution $u_n = An^2 + Bn + C$ does not work, since in this case, we get

$$0n^2 + (-2A)n + 5A - B = n^2 - 2n + 8$$

which cannot be correct for all n . We try

$$u_n = An^3 + Bn^2 + Cn + D.$$

Thus
$$u_{n-1} = A(n-1)^3 + B(n-1)^2 + C(n-1) + D,$$

$$u_{n-2} = A(n-2)^3 + B(n-2)^2 + C(n-2) + D,$$

and
$$u_n - 3u_{n-1} + 2u_{n-2} = -3An^2 + (15A - 2B)n - 13A + 5B - C.$$

Hence on comparing co-efficients we get

$$-3A = 1,$$

$$15A - 2B = -2,$$

$$-13A + 5B - C = 8,$$

which give $A = -1/3$, $B = -3/2$, $C = -67/6$. There is no condition on D , which means it can be chosen arbitrarily. The general solution of the equation can now be written as

$$y_n = a_1 + a_2 (2)^n - \frac{1}{3} n^3 - \frac{3}{2} n^2 - \frac{67}{6} n$$

where a_1 and a_2 are arbitrary constants. The arbitrary constant D has been absorbed in a_1 . The above examples illustrate that one should be careful in making a guess for the polynomial to be chosen as the particular solution of a given equation. However, if we write the equation in operator form, the form of the equation offers some help in this regard. To see this, we note that, if k is a positive integer and E is the shifting operator then

$$(E - 1)n^k = (n + 1)^k - n^k = kn^{k-1} + \dots$$

We see that $(E - 1)n^k$ is a polynomial of order $k-1$. If

$$P_k(n) = a_k n^k + a_{k-1} n^{k-1} + \dots + a_0,$$

it is clear that $(E - 1)P_k(n)$ will be a polynomial of degree $k-1$.

Also $(E - 1)^2 P_k(n)$ will be a polynomial of degree $k-2$ and so on.

The equation in Example 7.12 may be written as

$$(E^2 - 3E + 2)y_n = n^2 - 2n + 8,$$

or
$$(E - 2)(E - 1)y_n = n^2 - 2n + 8.$$

Now, if we were to take $u_n = An^2 + Bn + C$, $(E - 1)u_n$ must be a polynomial of degree 1, consequently $(E - 2)(E - 1)u_n$ will also be a polynomial of degree 1 and we cannot equate it with $n^2 - 2n + 8$ for all n . Since $E - 1$ reduces the degree of any polynomial in n by 1, we should assume

$$u_n = An^3 + Bn^2 + Cn.$$

Recently Martin [9] has discussed an interesting new method for finding the particular solution of a linear difference equation.

EXAMPLE 7.13: Find a particular solution of

$$(E - 1)^3(E + 3)y_n = 2n + 3.$$

SOLUTION: We should take u_n to be a polynomial of degree 4 i.e.

$$u_n = An^4 + Bn^3,$$

we stop at the n^3 term, because

$$(E - 1)^3(Cn^2 + Dn + F) \equiv 0,$$

and C , D and F drop out of the calculation.

EXAMPLE 7.14: Find a particular solution of

$$y_{n+2} - 3y_{n+1} + 2y_n = 4e^{2n}.$$

SOLUTION: We try $u_n = ce^{2n}$,

$$u_{n+1} = ce^{2}e^{2n}, \quad u_{n+2} = ce^4e^{2n}.$$

Thus u_n will satisfy the equation if

$$c(e^4 - 3e^2 + 2) = 4$$

or

$$c = 4/(e^4 - 3e^2 + 2)$$

and

$$u_n = 4e^{2n}/(e^4 - 3e^2 + 2).$$

EXAMPLE 7.15: Find a particular solution of

$$y_{n+2} + 4y_{n+1} + 3y_n = 2^n.$$

SOLUTION: We try $u_n = c2^n$, then

$$u_{n+1} = 2c2^n, \quad u_{n+2} = 4c2^n.$$

Hence u_n will satisfy the equation if

$$4c + 8c + 3c = 1 \quad \text{or} \quad c = 1/15.$$

Hence $u_n = (1/15)2^n$ is the particular solution of the equation.

EXAMPLE 7.16: Find a particular solution of

$$y_{n+2} + 4y_{n+1} + 3y_n = 4(-3)^n.$$

SOLUTION: Here the assumption $u_n = c(-3)^n$ fails because $(-3)^n$ happens to be a solution of the homogeneous equation

$$y_{n+2} + 4y_{n+1} + 3y_n = 0,$$

hence the substitution $u_n = c(-3)^n$ will naturally result in $0 = 4(-3)^n$ which does not determine c . We try

$$u_n = cn(-3)^n, \text{ then}$$

$$u_{n+1} = c(n+1)(-3)^{n+1} = -3c(n+1)(-3)^n,$$

and

$$u_{n+2} = 9c(n+2)(-3)^n.$$

So u_n will satisfy the equation if

$$\begin{aligned} & [9(n+2)c - 12c(n+1) + 3cn](-3)^n = 4(-3)^n \\ \text{or } & 18c - 12c = 4 \Rightarrow c = 2/3. \end{aligned}$$

Therefore $u_n = (2/3)n(-3)^n$ is the particular solution.

EXAMPLE 7.17: Find a particular solution of

$$y_n - 3y_{n-2} = \sin 2n.$$

SOLUTION: We try $u_n = A \sin 2n + B \cos 2n$

$$u_{n-2} = A \sin (2n - 4) + B \cos (2n - 4)$$

$$\begin{aligned} &= A [\sin 2n \cos 4 - \cos 2n \sin 4] \\ &+ B [\cos 2n \cos 4 + \sin 2n \sin 4]. \end{aligned}$$

Thus u_n satisfies the equation if

$$\begin{aligned} A - 3A \cos 4 - 3B \sin 4 &= 1, \quad 3A \sin 4 + B - 3B \cos 4 = 0, \\ \text{so } A &= (1 - 3 \cos 4)/(10 - 6 \cos 4), \quad B = -3 \sin 4/(10 - 6 \cos 4). \end{aligned}$$

EXAMPLE 7.18: Solve the problem

$$y_{n+2} - y_n = 2n^2, \quad y_0 = 1, \quad y_1 = 2.$$

SOLUTION: The characteristic equation is $\alpha^2 - 1 = 0$, with roots ± 1 .
To find the particular solution let

$$\begin{aligned} u_n &= An^3 + Bn^2 + Cn, \\ u_{n+2} &= A(n+2)^3 + B(n+2)^2 + C(n+2). \end{aligned}$$

We see that u_n will satisfy the equation if

$$A(n+2)^3 + B(n+2)^2 + C(n+2) - An^3 - Bn^2 - Cn = 2n^2 \text{ for all } n.$$

Or $6An^2 + (12A + 4B)n + 8A + 4B + 2C = 2n^2.$

This gives

$$\begin{aligned} 6A &= 2 &\Rightarrow A &= 1/3 \\ 12A + 4B &= 0 &\Rightarrow B &= -1 \\ 8A + 4B + 2C &= 0 &\Rightarrow C &= 2/3. \end{aligned}$$

Hence the general solution of the equation is

$$y_n = a + b(-1)^n + (1/3)n^3 - n^2 + (2/3)n$$

where a and b are arbitrary constant.

Now $y_0 = a + b \Rightarrow a + b = 1,$
 $y_1 = a - b \Rightarrow a - b = 2.$

Thus $a = 3/2$ and $b = -1/2.$

and we have the solution

$$y_n = 3/2 - (1/2)(-1)^n + (1/3)n^3 - n^2 + (2/3)n.$$

EXAMPLE 7.19: Find the solution of

$$y_{n+2} + 4y_n = 0, \quad y_0 = 2, y_1 = 6.$$

SOLUTION: The general solution in this case is

$$y_n = a 2^n \cos (\pi n/2) + b 2^n \sin (\pi n/2)$$

$$y_0 = a \Rightarrow a = 2$$

and

$$y_1 = 2b \Rightarrow b = 3.$$

The solution is

$$y_n = 2^{n+1} \cos (n\pi/2) + (3)2^n \sin (n\pi/2).$$

EXERCISE 7

1. Find the general solution of each of the following difference equations:

- (a) $y_{n+2} + 2y_{n+1} + 5y_n = 0$
- (b) $y_{n+2} - y_{n+1} - 2y_n = 0$
- (c) $y_{n+1} - 4y_n = 5$
- (d) $y_{n+2} - 4y_{n+1} + 4y_n = n^2 - 4$
- (e) $y_{n+3} - 2y_{n+2} - y_{n+1} + 2y_n = 4$
- (f) $y_{n+2} + y_n = \sin 2n$
- (g) $y_{n+2} + y_n = \cos n$
- (h) $y_{n+2} - y_{n+1} - y_n = 3^n$
- (i) $y_{n+2} + y_n = \sin(n\pi/2)$

2. Solve the following initial value problems:

- (a) $y_{n+2} - 3y_{n+1} + 2y_n = 4^n, \quad y_0 = 0, \quad y_1 = 1.$
- (b) $y_{n+2} - 3y_{n+1} + 2y_n = 2^n, \quad y_0 = 0, \quad y_1 = 2.$
- (c) $y_{n+2} - 4y_n = n^2, \quad y_0 = 1, \quad y_1 = 0.$
- (d) $\Delta y_n = 2y_n + 4, \quad y_0 = 1.$
- (e) $y_{n+2} - 4y_{n+1} + 4y_n = 2^{n+2}, \quad y_1 = 1, \quad y_2 = 2.$

3. $y_{n+2} = y_{n+1} + y_n, \quad y_0 = 0, \quad y_1 = 1.$

It is clear that the sequence $\{y_n\}_{n=0}^{\infty}$ is 0, 1, 1, 2, 3, 5, 8, 13,

These are called Fibonacci numbers. Solve the initial value problem.

4. Let
$$I_n = \int_0^1 \frac{x^n}{x^2 - x + 1} dx.$$

Show that $I_{n+2} - I_{n+1} + I_n = 1/(n+1)$. Find I_0 and I_1 directly, then calculate I_2, I_3, I_4 from the above difference equation.

5. Solve $(E - 2)(E - 1)^4 y_n = n^4 + 3n^2$.

6. If $I_n = \int_0^a x^n \sqrt{a^2 - x^2} dx$, then show that $I_{n+2} - \frac{n-1}{n+4} a^2 I_n = 0$.

Also, solve this difference equation.

7. Write the difference equation for

(a) $I_n = \int_0^{\pi/2} \cos^n x dx$, (b) $I_n = \int_0^{\pi/2} \sin^n x dx$,

(c) $I_n = \int_{\pi/4}^{\pi/2} \cot^n x dx$.

Solve the difference equation in each case.